Conjecture 00000000644: a Burau eigenvalue below the stated bound

lizaixi01, with OpenAI Codex assistance

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1 The explicit spectral condition

The source explicitly requires every eigenvalue of the Burau representation at real parameter points to have real part at least $-1/2$. Take the legal nonzero real parameter $t = 3/4$ and the generator σ_1 of the three-strand braid group. Its standard Burau matrix has the actual complex eigenvalue $-3/4$, whose real part is below $-1/2$. We give the reduced and unreduced versions, and prove the violation persists under every invertible change of basis.

This refutes the explicit necessary spectral clause of the stated conjunction. It does not refute an assertion merely about positivity of the ordinary character trace. It also does not construct or verify the proposed holomorphic-disk, LKB, or hyperbolic-volume mechanisms. Those additional structures cannot change an eigenvalue of the Burau representation by choosing a different basis.

2 A genuine braid-group representation

Use the standard Artin presentation

$$B_3 = \langle \sigma_1, \sigma_2 \mid \sigma_1\sigma_2\sigma_1 = \sigma_2\sigma_1\sigma_2 \rangle.$$

Morier-Genoud, Ovsienko and Veselov, *Burau representation of braid groups and q-rationals*, p.1, equation (1.1), define the reduced Burau representation on these generators by

$$A(t) = \begin{pmatrix} -t & 1 \\ 0 & 1 \end{pmatrix}, \qquad B(t) = \begin{pmatrix} 1 & 0 \\ t & -t \end{pmatrix}.$$

Their p.2 defines specialization at every nonzero complex t, including nonzero real parameters. Lean implements these very matrices, proves both determinants equal $-t$, and proves $ABA = BAB$ for all complex t. For $t \neq 0$ they are actual elements of $\mathrm{GL}_2(\mathbf{C})$. The universal property of the presented group then constructs the group homomorphism and proves its values on σ_1. This is not an isolated matrix assigned a representation's name.

For the standard unreduced version, use

$$U(t) = \begin{pmatrix} 1-t & t & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \qquad V(t) = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1-t & t \\ 0 & 1 & 0 \end{pmatrix}.$$

These are the generator blocks in Kassel and Turaev, *Braid Groups*, Section 3.1.1, pp.93–94. Lean separately proves their determinants, braid relation, actual invertibility, induced group homomorphism, and generator value.

3 Actual eigenvectors and arbitrary bases

At $t = 3/4$, the nonzero vectors

$$e = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \qquad u = \begin{pmatrix} -3 \\ 4 \\ 0 \end{pmatrix}$$

satisfy

$$A(3/4)e = -\frac{3}{4}e, \qquad U(3/4)u = -\frac{3}{4}u.$$

Lean proves both vector equations and both nonzeroness conditions, then uses Mathlib's actual `Module.End.HasEigenvalue` for the complex linear maps induced by these matrices. The result is a complex eigenvalue statement, not just an unconnected scalar inequality. It also proves that the parameter is nonzero, has zero imaginary part, and lies strictly between zero and one on the real axis, and that

$$\mathrm{Re}(-3/4) = -3/4 < -1/2.$$

For any invertible complex matrix P, if $Mv = \mu v$ and $v \neq 0$, put $w = P^{-1}v$. Invertibility implies $w \neq 0$, and

$$(P^{-1}MP)w = P^{-1}Mv = \mu P^{-1}v = \mu w.$$

The general Lean theorem `conjugate_eigenvalue` proves this using actual invertible matrices and nonzero vectors. Applying it in dimensions two and three shows that every conjugate of the respective σ_1 matrix violates the requested real-part lower bound. Thus a proposed Hill basis, if it is a basis of the same representation, cannot repair this spectral violation. No definition of that unspecified basis is needed to prove the stronger assertion for every basis.

4 Formal scope and verification

`Braid3` is Mathlib's `PresentedGroup` on two generators with the genuine Artin relator. Both Burau specializations are actual group homomorphisms to the relevant general linear group. All specialization and relation checks are formal, with no assumptions about matrices satisfying the braid relation.

`SpectralBound` quantifies over every actual complex eigenvalue of the matrix's complex linear map and imposes exactly the source's real-part inequality. The two basis-independent counterexample results are followed by closed theorems negating the universal real-specialization spectral condition in both standard versions. The parameter and braid are legitimate instances of the original unrestricted spectral clause. No restriction to a different braid class or parameter interval is inserted.

Lean 4.33.0 and Mathlib commit `db584cd6d46c92f209a44c0f1c829460d327499d` are pinned. The submission includes a new Main build, twenty-three transitive axiom audits, and full official `leanchecker --fresh Main` replay with source-bound commands, exit codes and logs. No `sorry`, `native_decide` or added axiom is used. OpenAI Codex assisted; a separate supervising Codex agent reviewed the formal chain and report. This is AI review, not human certification. Organizer acceptance is pending.